MIL-F-8193(USAF)
7 November 1952

MILITARY SPECIFICATION

FELT, WOOL, COTTON, KAPOK AND COMBINATION THEREOF

1. SCOPE

1.1 Scope.- This specification covers 3 types of insulation felt.

1.2 Classification.- The kapok blended felt shall be of the following types and forms, as specified by the procuring agency: (see 6.2)

Type I - Composed of Wool, Cotton and Kapok Fibers.
Type II - Composed of Wool and Kapok.
Type III - Kapok Principal Fiber.

Type III shall be furnished in the following forms, as specified: (see 6.2)

Flat
Corrugated.

2. APPLICABLE SPECIFICATION, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specification of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

CCC-T-191	Textile Test Methods

MILITARY

MIL-P-6065	Packaging, Packing and Marking of Cloth, Fabric and Webbing.

(Copies of specifications and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials

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3.1.1 Wool.- The wool fiber used in the felt shall be of a good grade and of sufficient staple length to meet the requirements of this specification.

3.1.2 Cotton.- The cotton fibers shall be of a grade and staple length suitable for the production of a good felt when processed with wool and kapok.

3.1.3 Kapok.- The kapok shall be of a suitable grade to process with wool and cotton so that the finished felt shall conform to the requirements of this specification.

3.2 Composition of types:

3.2.1 Type I felt.- The felt shall contain fibers of approximately the following proportions:

Cotton - 33 percent
Wool - 27 percent
Kapok - 40 percent.

3.2.2 Type II felt.- The felt shall contain kapok as the principal fiber and wool may be added in sufficient quantities to insure the manufacture of suitable materials.

3.2.3 Type III felt.- The felt shall contain approximately 80 percent of kapok fibers and the remainder shall consist of commercially acceptable fibers and binders to insure the manufacture of suitable material.

3.3 General requirements

3.3.1 Width.- Unless otherwise specified, the materials shall be 36 inches wide with a tolerance of plus or minus 1/2 inch.

3.3.2 Length.- The length of cuts of the material shall be as specified by the procuring agency. (see 6.2) A tolerance of plus or minus one percent will be allowed.

3.3.3 Color.- The color of the felt shall be natural (greige).

3.3.4 Flameproofing.- The material shall be given a flameproofing treatment using commercially accepted flame retarders. No materials shall be used for this purpose that will cause deterioration of the fibers or render the finished felt unsuitable for the intended use. The flame proofing treatment shall be such as to meet the test for flame proofness of 4.7.2

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3.4 Detail requirements

3.4.1 Type I and type II felts shall conform to the following thickness and density requirements: (see 6.2)

Type I		Type II	
Thickness (in.) ± 1/16	Weight (lbs/sq.yd.) ± 10%	Thickness (in.) ± 1/16	Weight (lbs/sq.yd) ± 10%
1/8	0.55		
1/4	1.09	1/4	1.14
3/8	1.53		
1/2	2.18	1/2	2.88
5/8	2.52		
3/4	3.11		
7/8	3.53		
1	4.32		

3.4.2 Type III felts, furnished in flat and corrugated forms, shall have the following physical requirements: (see 6.2)

Flat		Corrugated			
Thickness (in.) ± 1/16	Weight (lbs/sq.yd) ± 10%	Thickness between corrugations ± 1/16	Thickness over corrugations ± 1/16	Weight (lbs/sq yd) ± 10%	Corrugations per 6 inches
1/8	0.45	3/16	5/8	1.45	6-7

35. Workmanship.- The finished felts shall be well felted from uniformly mixed materials and shall be uniform in thickness and free from holes, tears, lumps, wrinkles, foreign matter or any other imperfections which would affect the serviceability or appearance of the material.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection tests.- All the tests specified herein are classified as inspection tests. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government inspector. Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or material shall not relieve the contractor of responsibility for faulty workmanship or material which may be discovered at any time prior to final acceptance, and shall in no case be construed as a guaranty of the acceptance of the finished product. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable.

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4.2 Inspection.- Inspection shall consist of visual inspection and all tests specified herein. Inspection shall be at point of manufacture unless otherwise specified by the procuring agency. (see 6.2) In case of inspection at factory, every facility for the prosecution of this work shall be afforded by the manufacturer to the inspector.

4.2.1 Visual inspection.- Visual inspection shall be 100 percent and shall include the inspection of the packaging, packing, and marking, and of conformance to requirements of this specification not covered by tests. Visual inspection for defects in the felt shall be accomplished before the material is prepared for shipment.

4.3 Rejection and retest.- Failure of any sample of felt to conform to any of the requirements of this specification shall be cause for rejection of the lot represented.

4.3.1 Resubmitted lots.- When the supplier elects to resubmit a rejected lot, the supplier shall first inspect the rejected lot, repair or remove all defects such as would otherwise render the lot rejectable, and then present the lot for reinspection.

4.3.2 Conditions of resubmission.- Felt which has been rejected may be resubmitted after correction of the condition, or conditions, which caused rejection. Before resubmitting, full particulars concerning previous rejection and the action taken to correct defects found in the prior inspection shall be furnished to the inspector. Material rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.4 Disposition of rejected product.- The Government reserves the right to return to the supplier any rolls found defective in the course of, and by reason of visual inspection, whether or not the lot as a whole is accepted.

4.5 Sample for physical testing.- A sample, or samples, each at least 2 yards long and full width of the felt, to be furnished at no additional cost to the Government, shall be taken from each 500 yards submitted for acceptance on any contract or order. Each sample shall be suitably tagged to identify it with the roll sampled and such samples shall be subjected to all tests specified herein. When an inspection lot exceeds 500 yards, an additional sample for each 500 yards or fraction thereof of the excess shall be taken.

4.6 Atmospheric conditions for testing.- Unless otherwise specified in the test method, all tests shall be made on specimens in standard condition as described in Specification CCC-T-191, Section 4.

4.7 Tests.- The methods of testing specified in Specification CCC-T-191, except as modified in 4.7.1, 4.7.2, and 4.7.3 of this specification, shall be followed, wherever applicable.

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4.7.1 Weight.- The density determination shall be accomplished on a piece of representative material, not less than 12 inches square.

4.7.2 Flameproofness.- A specimen not less than six inches square shall be held for 30 seconds over the tip of a blue Bunsen flame which is adjusted to a height of 1-3/4 inches. The specimen shall be removed and no flame shall propagate from its surface. This test shall be accomplished under a hood with the fan idle to provide for a minimum movement of air currents.

4.7.3 Thickness.- The thickness shall be measured by laying the specimen on a flat surface and placing a steel rule or scale approximately 12 inches long by one inch wide by 1/16 inch thick, along the edge of the specimen. The thickness shall be measured between the surface and the underside of the steel rule or scale.

5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging,and packing.- The felt shall be packaged and packed in accordance with the requirements of Specification MIL-P-6065.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked to conform with the requirements of Specification MIL-P-6065.

6. NOTES

6.1 Intended use.- The felts are intended for use in aircraft and aircraft maintenance equipment such as air vent plugs in hydraulically operated equipment and miscellaneous packing and insulating parts.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Type and form required (see 1.2)
c. Thickness required (see 3.3)
d. Length of cuts (see 3.3)
e. Point of inspection

6.3 Supersedure note.- This specification provides material similar to that formerly procured under Air Force Specification No. 16098A, Felt; Wool, Cotton, Kapok and Combination Thereof, which is cancelled.

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